

WhatchyaDoin?

Wanting, Starting, and Noticing Are Different Questions

Human-readable companion to “WhatchyaDoin? Separating Intention, Action Realization, Causal Provenance, and Awareness”

Abstract

You can want to do something, know how to do it, and still not start. When you finally begin, the action itself does not reveal which internal change allowed it to happen. And when you notice yourself acting, that awareness is not automatically evidence that it initiated the action. WhatchyaDoin? keeps four evidential questions separate: did the person intend the action, did that particular action become behaviorally realized, what caused that realization, and what was noticed or reported about it? The distinctions are established across action science; the paper’s proposed contribution is a common identification audit that prevents evidence for one question from being substituted for another. It supplies no new action mechanism and no universal timeline for consciousness.

1. The question that changes what you notice

You are reading this sentence. Now ask yourself: “WhatchyaDoin?” You may suddenly recognize what was already underway: “I am reading.” But that moment of explicit recognition does not tell you whether you were aware in some other sense a moment earlier. It also does not tell you what caused the reading to begin. The question makes one relation visible without explaining every relation behind it.

The more difficult version occurs before action starts. You mean to answer a message, begin a document, get out of the chair, or take a walk. The action is familiar. The intention may be genuine. Yet nothing happens. Later, perhaps with no new instruction about what to do, you start. What changed? The temptation is to infer a hidden switch from the outcome. The paper instead asks what can actually be identified.

2. Four different kinds of evidence

WhatchyaDoin? treats the ordinary sequence “I wanted to, I started, I noticed” as several questions rather than one causal story. The notation is bookkeeping, not a proposal for four brain modules.

Question	Scientific target	What the observation does NOT settle
Did I want or intend to act?	W: independently measured wanting or intending relevant to action A.	Whether the action became realized. Non-action does not establish that W was absent.
Did that action start?	R_A: the predeclared, action-specific behavioral realization criterion.	Which internal mechanism caused the onset or whether an intention had sole causal authority.
What caused the onset?	Causal provenance of R_A, tested through suitable interventions and competing-route measures.	Not determined by R_A itself or a persuasive after-the-fact explanation.
When and what did I notice?	N_A: a locally defined awareness measure about action A, distinct from agency feelings and judgments.	Whether awareness caused the action, when every kind of consciousness first existed, or whether a later report is a transparent record.

For typing, R_A might be the first task-directed keystroke or a short matching sequence. For standing, it might be a defined postural transition. For sending a message, the relevant criterion could be beginning composition or pressing Send. The definition belongs to the question being tested. It cannot simply be “the moment the intention gained control,” because that would build the proposed cause into the outcome being measured.

3. Wanting and knowing are not the same as beginning

Intention–behavior research already shows that stronger intentions do not produce a one-for-one increase in action (Webb & Sheeran, 2006). Implementation intentions and action-control work show that cue-response preparation and self-regulatory conditions can affect enactment (Gollwitzer & Sheeran, 2006; Koole et al., 2023). None establishes a universal missing stage between wanting and doing.

Capability introduces another distinction. A person may perform an action when externally cued even though the self-paced version is difficult. Self-initiated versus externally triggered movement research provides a clear precedent. But successful cued execution does not prove exactly why self-initiation previously failed, and later competent performance does not prove that every earlier enabling condition was present. To understand the gap, initiation and post-onset execution need separate measures.

4. Same first keystroke, different causal stories

The manuscript's worked example is deliberately ordinary. A participant learns a target phrase. The phrase, keyboard, posture, and instructions stay the same. They are asked to begin typing when ready. The first valid task-directed keystroke crosses R_A. The experiment now knows that typing began at that moment.

What it does not know is why. The person may have held a plan that became executable as readiness changed, responded to a learned cue, retrieved or reconstructed part of the phrase, shifted attention toward the task, resolved competition with another activity, or changed a control policy. Each route could produce the same keystroke at the same latency and an equally correct completed phrase. The endpoint alone cannot select among those explanations.

A randomized cue or support manipulation may show that onset probability or latency changes. That is a real causal result about the manipulated condition. It is not automatically a causal result about the particular latent route. More selective intervention, independent representation measures, or explicit alternative-model comparisons are needed to narrow the provenance claim.

5. No new instruction is not the same as no new internal content

This boundary is easy to miss. Suppose the target phrase and instructions remain unchanged while a change in task context makes typing more likely. Three statements sound similar but are not:

- No new EXTERNAL action identity or instruction was supplied. This can often be secured from the design.
- No new LATENT INTERNAL action-specific content was formed. This needs independent evidence; retrieval or replanning may still occur.
- Nothing internally changed. This is not the manuscript's claim. A changed onset must have some causal difference, even if it is not a new action description.

The interrupt-request analogy helps only in a restricted sense: a transition-enabling signal need not itself contain the full action it helps release. Dopamine movement-initiation experiments and proposals about noradrenergic reconfiguration illustrate possible architectural roles, not one universal brain interrupt or a dopamine/norepinephrine command to act (da Silva et al., 2018; Bouret & Sara, 2005). The relevant question is which process changed and what evidence distinguishes it—not which label best fits the observed start.

6. Noticing is neither the action nor its origin

The word “awareness” covers different measurements: conscious intention before movement, awareness of movement, explicit recognition of an ongoing activity, an online report prompted during the episode, a retrospective account, a feeling of agency, and a judgment of agency. These are not interchangeable. The paper's N_A is only the specific construct a study operationally defines and validates.

Research on conscious-intention timing, retrospective judgments, and motor awareness already shows why a simple “brain decides first, consciousness notices later” rule cannot be treated as universal. But rejecting that sequence does not prove the reverse rule that consciousness always initiates action. A failed report probe also does not prove an absence of phenomenal experience. It shows a failure to obtain that report under that measurement method.

Explicit noticing may nevertheless affect what happens next. After recognizing that you are reading, you might slow down, reread, stop, or change direction. But asking “WhatchyaDoin?” also alters attention, demand, expectancy, and strategy. A prompt followed by changed behavior is evidence for the broader prompt or monitoring condition—not for awareness-specific causation unless those alternative pathways have been independently constrained.

7. What a discriminating experiment would have to test

The paper supplies five complementary test families. None is a universal experiment, and each can narrow some explanations without identifying a unique mechanism:

Test family	Discriminating design question
Intention vs. realization	Can action support change R_A while independently measured W remains approximately stable?
Capability vs. self-initiation	Does a cue help onset more than post-onset execution of a comparable action?
Content vs. transition	Can externally fixed action identity and independent content assays constrain retrieval/replanning alternatives as transition support changes?
Realization vs. awareness	Can R_A and a locally validated N_A vary separately without using one to define the other?
Awareness as downstream input	Does an awareness-specific effect survive controls for attention, expectancy, feedback, strategy, stimulation, and report reactivity?

If an awareness manipulation cannot be separated from the other processes it changes, the correct conclusion is narrower: the monitoring condition affected continuation, correction, or stopping. The awareness-specific mediator remains unidentified. Likewise, constant external action identity is not enough to claim invariant internal content.

8. The five-question audit and when to stop

The manuscript condenses its approach to five questions that apply whether the experiment concerns typing, reaching, reading, or some other declared action:

- What was actually observed?
- Which relevant construct was measured independently rather than inferred from that observation?
- What was experimentally manipulated?
- Which latent explanations still predict the same result?
- Which conclusion did the evidence discriminate, and which stronger claims remain unsupported?

The distinctions should contract where the task cannot meaningfully separate intention, realization, and awareness. A content-versus-transition claim fails to be identified if content changes inseparably with the manipulation. An awareness-specific causal claim remains open if conscious noticing cannot be distinguished from attention, feedback, expectancy, or task-demand effects. No measurement gap should be repaired with a theory-specific name for an unobserved stage.

9. What WhatchyaDoin? adds—and what it does not

Action-control theory, implementation intentions, What–When–Whether models, phenomenology of action, voluntary-movement research, and sense-of-agency accounts already own many of the individual distinctions. This paper does not claim to discover those components or to supply a new neural pathway. Its possible contribution is their disciplined composition: one operational evidence audit across intention, action-specific realization, provenance, and awareness. The manuscript keeps exact paper-level originality open; materially equivalent prior cross-literature work would reduce or subsume that contribution.

The question in the title returns with a narrower and more useful meaning. “WhatchyaDoin?” can make the action visible to the person doing it. It cannot, by itself, reveal what first allowed the action to happen. Wanting, knowing, starting, and noticing are different facts. An explanation is earned by measuring the relation it claims, not by reading an entire causal story backward from one successful action.

Selected references

- Banks, W. P., & Isham, E. A. (2009). We infer rather than perceive the moment we decided to act. *Psychological Science*, 20(1), 17–21.
- Bouret, S., & Sara, S. J. (2005). Network reset: A simplified overarching theory of locus coeruleus noradrenaline function. *Trends in Neurosciences*, 28(11), 574–582.
- Brass, M., & Haggard, P. (2008). The What, When, Whether Model of Intentional Action. *The Neuroscientist*, 14(4), 319–325.
- Chambon, V., Sidarus, N., & Haggard, P. (2014). From action intentions to action effects: How does the sense of agency come about? *Frontiers in Human Neuroscience*, 8, 320.
- da Silva, J. A., Tecuapetla, F., Paixao, V., & Costa, R. M. (2018). Dopamine neuron activity before action initiation gates and invigorates future movements. *Nature*, 554, 244–248.
- Gollwitzer, P. M., & Sheeran, P. (2006). Implementation intentions and goal achievement: A meta-analysis of effects and processes. *Advances in Experimental Social Psychology*, 38, 69–119.
- Koole, S. L., Jostmann, N. B., & Baumann, N. (2023). From intentions to action: An integrative review of action control theory and research. *Advances in Experimental Social Psychology*, 68, 291–375.
- Pacherie, E. (2008). The phenomenology of action: A conceptual framework. *Cognition*, 107(1), 179–217.
- Synofzik, M., Vosgerau, G., & Newen, A. (2008). Beyond the comparator model: A multifactorial two-step account of agency. *Consciousness and Cognition*, 17(1), 219–239.
- Webb, T. L., & Sheeran, P. (2006). Does changing behavioral intentions engender behavior change? A meta-analysis of the experimental evidence. *Psychological Bulletin*, 132(2), 249–268.